

In-Class Lab: A/B Testing on a Marketing Campaign

Lab Overview:

This lab will guide you through the process of conducting an A/B test using a real-world dataset. You will formulate hypotheses, perform statistical tests, and interpret results. The dataset used simulates data from a marketing campaign designed to improve a website's conversion rate by testing two different homepage designs.

Objective:

By the end of this lab, you will:

1. Learn the principles of A/B testing.
 2. Apply statistical methods (two-proportion z-test) to compare conversion rates.
 3. Analyze and interpret the results.
 4. Use Python to conduct hypothesis testing on a real-world dataset.
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Dataset:

- **Source:** Simulated dataset mimicking real-world marketing A/B testing data.
- **Dataset URL:** [Kaggle - A/B Testing Data](#)
- **Key Variables:**
 - `user_id`: Unique identifier for each user.
 - `group`: Group the user was assigned to (Control or Test).
 - `landing_page`: The version of the landing page the user saw (Old Page vs. New Page).
 - `converted`: Whether the user converted (1 = yes, 0 = no).

- `timestamp` : When the user visited the site.
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Lab Instructions:

Part 1: Data Exploration and Preparation

Step 1: Load the Data

- Load the provided dataset and inspect the first few rows to understand its structure.
- Check for any missing values in the dataset.
- Verify the data types of each column.

Step 2: Data Cleaning

- Identify if there are any duplicate users (i.e., users who appear in both the Control and Test groups).
- If duplicates are found, remove them from the dataset to ensure the test's integrity.

Step 3: Descriptive Statistics

- Calculate basic summary statistics (e.g., mean, median) for conversion rates, page views, and session time.
 - Group the data by landing page and display the average conversion rate for each group.
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Part 2: Hypothesis Formulation

Step 1: Define Hypotheses

- Based on the dataset, define your null and alternative hypotheses.
 - **Null Hypothesis (H_0):** The new landing page has no effect on conversion rates (i.e., conversion rates are the same for both the Test and Control groups).

- **Alternative Hypothesis (H_1):** The new landing page increases conversion rates.

Step 2: Data Visualization

- Create a bar plot that visualizes the conversion rates for the Test and Control groups.
 - Interpret the differences (if any) in conversion rates between the two groups.
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Part 3: A/B Testing (Two-Proportion Z-Test)

Step 1: Perform a Two-Proportion Z-Test

- Using the dataset, calculate the conversion rates for both the Control and Test groups.
- Perform a two-proportion z-test to determine whether the difference in conversion rates between the Control and Test groups is statistically significant.
- Record the z-statistic and p-value from the test.

Step 2: Interpretation

- Based on the p-value, decide whether to reject or fail to reject the null hypothesis.
 - Discuss the business implications of your findings—should the company switch to the new landing page?
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Part 4: Confidence Intervals and Visualization

Step 1: Confidence Intervals

- Calculate the 95% confidence intervals for the conversion rates of both the Test and Control groups.
- Discuss what the confidence intervals indicate about the reliability of the results.

Step 2: Visualization of Confidence Intervals

- Create a plot that shows the conversion rates for both groups, including the confidence intervals as error bars.
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Part 5: Subgroup Analysis

Step 1: Time-Based Analysis

- Using the timestamp data, extract the hour of the day that each user visited the website.
- Group the data by hour and calculate the conversion rates for each hour.
- Create a line plot that shows how conversion rates vary by the time of day.

Step 2: Interpretation

- Analyze the conversion rate trends over different times of day.
 - Identify if there are any specific time periods where the new landing page was more or less effective.
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Part 6: Conclusion

Step 1: Summarize Findings

- Summarize the results of your A/B test, including whether the new landing page showed a statistically significant improvement in conversion rates.
- Discuss any additional insights gained from the confidence intervals and subgroup analysis.

Step 2: Recommendations

- Based on your analysis, recommend whether the company should adopt the new landing page.
 - Highlight any potential limitations of the A/B test, such as sample size, duration of the test, or external factors.
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Lab Deliverables

1. **Data Exploration:** Ensure the dataset is clean and present descriptive statistics.
 2. **Hypothesis Formulation:** Write the null and alternative hypotheses and explain why you selected them.
 3. **A/B Test Results:** Present the results of the two-proportion z-test and interpret the p-value.
 4. **Confidence Intervals:** Calculate and plot 95% confidence intervals for conversion rates.
 5. **Subgroup Analysis:** Perform an additional analysis based on time of day and provide insights.
 6. **Final Report:** Write a report summarizing the entire lab, including your findings, visualizations, and recommendations for the company.
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Learning Outcomes:

By the end of this lab, you should be able to:

- Understand and apply A/B testing principles to a real-world scenario.
 - Perform hypothesis testing using the two-proportion z-test.
 - Calculate confidence intervals and interpret their meaning.
 - Conduct further analysis on subgroups of data (e.g., time-based analysis).
 - Draw actionable insights from your analysis to inform business decisions.
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